

Chapter 9

Squeeze Theorem for Sequences. Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ with $a_n \leq b_n \leq c_n$ for all integers n greater than some index N . If $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = L$, then $\lim_{n \rightarrow \infty} c_n = L$.

Growth Rate of Sequences. The following sequences are ordered according to increasing growth rates as $n \rightarrow \infty$: $\{\ln n\} \ll \{n^p\} \ll \{n^p \ln n\} \ll \{n^{p+s}\} \ll \{b^n\} \ll \{n!\} \ll \{n^n\}$. The ordering applies for positive real numbers p, r , and s and $b > 1$.

Geometric Series. Let $r, a \in \mathbb{R}$. If $|r| < 1$, then $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$. If $|r| \geq 1$, then the series diverges.

Divergence Test. If $\sum a_k$ converges, then $\lim_{n \rightarrow \infty} a_k = 0$. Equivalently, if $\lim_{n \rightarrow \infty} a_k \neq 0$, then the series diverges.

Integral Test. Suppose f is a continuous, positive, decreasing function for $x \geq 1$ and let $a_k = f(k)$ for $k = 1, 2, 3, \dots$. Then

$$\sum_{k=1}^{\infty} a_k \text{ and } \int_1^{\infty} f(x) dx$$

either both converge or both diverge. In the case of convergence, the value of the integral is *not*, in general, equal to the value of the series.

Chapter 10

Taylor's Polynomials. Let f be a function with $f', f'', \dots, f^{(n)}$ defined at a . The **n th-order Taylor Polynomial** for f with its **center** at a denoted p_n , has the property that it matches f in value, slope, and all derivatives up to the n th derivative at a ; that is,

$$p_n(a) = f(a) \text{ and } p'_n(a) = f'(a), \dots, p_n^{(n)}(a) = f^{(n)}(a).$$

The n th-order Taylor polynomial centered at a is

$$p_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

More compactly,

$$p_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k \text{ for } k = 0, 1, 2, \dots, n.$$

Helpful formulas

$$1. \frac{d}{dx} (x^n) = nx^{n-1}$$

$$2. \frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x) \text{ (Chain rule)}$$

$$3. \frac{d}{dx} (f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

$$4. \frac{d}{dx} e^x = e^x$$

$$5. \frac{d}{dx} \sin ax = a \cos ax$$

$$6. \frac{d}{dx} \cos ax = -a \sin ax$$

$$7. \int x^n dx = \frac{1}{n+1} x^{n+1} + C; n \neq -1$$

$$8. \int \cos ax dx = \frac{1}{a} \sin ax + C$$

$$9. \int \sin ax dx = -\frac{1}{a} \cos ax + C$$

$$10. \int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

$$11. \int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$$